

Question 1 answer: In C#, array elements are initialized to default values (e.g., 0 for numeric types, false for bool, null for reference types).

Question 2 answer: `Array.Clone()` creates a shallow copy of the array, while `Array.Copy()` copies elements to another existing array.

Question 3 answer: `GetLength(dimension)` returns the size of a specific dimension; `Length` returns the total number of elements in all dimensions.

Question 4 answer: `Array.ConstrainedCopy()` throws exceptions on failure and guarantees rollback, while `Array.Copy()` may leave data partially copied if it fails.

Question 5 answer: `foreach` is preferred for read-only because it prevents accidental modification and simplifies syntax.

Question 6 answer: Input validation prevents invalid, harmful, or unexpected input, enhancing security and reliability.

Question 7 answer: Format a 2D array using nested loops with consistent spacing or tabs for aligned and readable output.

Question 8 answer: Use `switch` when checking a single variable against many constant values for better readability and performance.

Question 9 answer: `Array.Sort()` has average time complexity $O(n \log n)$.

Question 10 answer: `for` is slightly more efficient because it avoids the overhead of the `foreach` iterator.